SQL Learning Journal #5 – Combining Filtering, Counting and Missing Values

Introduction

In today's lesson, I learned how to combine SQL filtering and counting functions in a single query. I also studied how SQL handles missing values (NULL values) and the differences between various COUNT functions.

1. Combining Multiple Conditions

```
SELECT COUNT(DISTINCT people_name) AS unique_people_name
FROM people_name_table
WHERE age BETWEEN 10 AND 19
AND language = 'French';
```

This query counts unique people aged between 10 and 19 who speak French.

2. Counting Missing and Non-Missing Values

COUNT(*) counts all rows including NULL rows.
COUNT(field_name) counts only non-NULL values.
COUNT(DISTINCT field_name) counts unique non-NULL values.

3. Using IS NULL and IS NOT NULL

```
SELECT COUNT(*) AS count_no_age FROM table_name WHERE field_name_age IS NULL;

SELECT COUNT(field_name_age) AS count_age FROM table_name WHERE
field_name_age IS NOT NULL;
```

COUNT(age) automatically ignores NULL values, so WHERE age IS NOT NULL returns the same count.

What I Learned Today

Combining filtering conditions with counting functions makes SQL queries more powerful. Regular practice is essential to remember the syntax and write queries confidently.